Estimate quotients using multiples

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Estimate and then calculate 7,392 / 24. Copy or complete the first step.

2. Estimate 5,985 / 31.

3. Which is a better estimate for 8,316 / 42: 20, 200 or 2,000?

4. Miro says: Miro estimates division by rounding only the dividend. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Estimate and then calculate 7,392 / 24. Copy or complete the first step.
   Answer About 300; exactly 308.

2. Estimate 5,985 / 31.
   Answer About 200 because 6,000 / 30 is 200.

3. Which is a better estimate for 8,316 / 42: 20, 200 or 2,000?
   Answer 200.

4. Miro says: Miro estimates division by rounding only the dividend. Is this correct? Explain.
   Answer Round both numbers to compatible values that are easy to divide.

Estimate quotients using multiples

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ____________

1. Estimate 5,985 / 31.

2. Which is a better estimate for 8,316 / 42: 20, 200 or 2,000?

3. Find a divisor between 20 and 30 that makes 6,048 divide exactly.

4. Identify and correct this misconception: Miro estimates division by rounding only the dividend.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Estimate 5,985 / 31.
   Answer About 200 because 6,000 / 30 is 200.

2. Which is a better estimate for 8,316 / 42: 20, 200 or 2,000?
   Answer 200.

3. Find a divisor between 20 and 30 that makes 6,048 divide exactly.
   Answer Possible answers include 21, 24 or 28.

4. Identify and correct this misconception: Miro estimates division by rounding only the dividend.
   Answer Round both numbers to compatible values that are easy to divide.

5. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Estimate quotients using multiples

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Find a divisor between 20 and 30 that makes 6,048 divide exactly.

2. Explain why this reasoning fails: Miro estimates division by rounding only the dividend.

3. Create a new example that tests the same idea as: Which is a better estimate for 8,316 / 42: 20, 200 or 2,000?

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Find a divisor between 20 and 30 that makes 6,048 divide exactly.
   Answer Possible answers include 21, 24 or 28.

2. Explain why this reasoning fails: Miro estimates division by rounding only the dividend.
   Answer Round both numbers to compatible values that are easy to divide.

3. Create a new example that tests the same idea as: Which is a better estimate for 8,316 / 42: 20, 200 or 2,000?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

4. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

5. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.